E-RICKSHAW REVOL

India is witnessing a boom in electric rickshaws. While these must transition to safer products, their bigger sibling—the electric auto—needs a leg-up with better charging infrastructure

By Ajay Sukumaran



Over a decade ago, New Delhi welcomed battery-operated e-rickshaws ahead of the 2010 Commonwealth Games as an eco-friendly alternative for mobility in a city battling air pollution. Since then, these low-speed three-wheelers, which ferry passengers over short distances for as little as Rs 10, have become ubiquitous, not just in Delhi but across the entire swathe of north India up to Assam in the east.

In 2022, e-rickshaws notched up a milestone of sorts—for the first time, they outsold the regular autorickshaws with internal combustion engines (ICE) running on fuels like CNG and LPG. According to data from the Federation of Automobile Dealers Associations (FADA), 294,000 e-rickshaws were sold last year as against 246,000 ICE autorickshaws. Driving this surge is the need for last-mile connectivity that has now gone beyond the big cities and into tier 2 and 3 towns, says Sudip Banerjee, the head of strategy and planning at the Delhi-based Saera Electric Auto, one of the earliest manufacturers of e-rickshaws in India. "Post-Covid, there has been a boom," he says. Riding this boom is a whole ecosystem of manufacturers, fleet operators, finance companies and start-ups. With e-rickshaws leading the charge, the three-wheeler industry—for long the workhorse of urban and rural mobility—is getting an electric makeover.

It's playing out in several ways. The basic e-rickshaws, modelled on the old cycle-rickshaws, run on lead-acid batteries. Up until January 2015, when the Union government brought e-rickshaws under the Motor Vehicles Act via an amendment, they had been operating in a grey area. "It was not mainstream even five years ago," says Pulkit Khurana, co-founder of the Delhi-based Battery Smart, which operates a Lithium-ion (Li-ion) battery-swapping network. But after they were legalised—which meant they could be registered with number plates like other vehicles—the e-rickshaws began to proliferate. There are over 400